

- 5 (a) Starting from the definitions of cosh and sinh in terms of exponentials, prove that

$$2 \cosh^2 x = \cosh 2x + 1. \quad [3]$$

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- (b) Find the solution of the differential equation

$$\frac{dy}{dx} + 2y \tanh x = 1$$

for which $y = 1$ when $x = 0$. Give your answer in the form $y = f(x)$. [8]

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